Adaptive Sensing beyond Non-Adaptive Information Bounds: End-to-End Co-Design of Hardware, Policy, and Inference

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Abstract

Inverse design has made vast physical parameter spaces a substrate for emergent behavior. In sensing, the stakes of this principle are sharpest at the analog-to-digital boundary, where any information the hardware fails to capture is information no downstream algorithm can recover; hardware optimization alone is therefore not enough, and the geometry must be co-designed with a rule for what to measure next. We formulate this co-design as *joint dynamic programming* (joint-DP): a single optimization over the continuous hardware $\boldsymbol{c}$ and a Bellman-optimal adaptive measurement policy π, made tractable by an envelope-theorem outer gradient that removes the policy Jacobian from the outer loop and by a relaxation hierarchy that carries a common framework from small discrete POMDPs to 10^5-pixel photonic topologies. Joint-DP strictly improves on the two natural baselines that historically design only half of the problem. On a radar beam-search POMDP, classical information-bound-guided geometry selection loses 2.8× in attainable adaptive value. On a superconducting-qubit flux sensor, joint-DP reduces deployed mean-squared error by 11.3× over the joint Bayesian Cramér–Rao baseline, with both hardware geometries numerically co-optimized under their respective objectives. A 90,000-pixel photonic metasensor co-designed via the Bayesian Fisher-information-matrix surrogate achieves a 123× reduction in deployed mean-squared error relative to a non-co-designed baseline. For any sensor whose hardware is designed once but whose policy runs forever, joint $(\boldsymbol{c}, \pi)$ optimization is the minimum principled procedure.

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1 Introduction

Modern deep learning has revealed that emergent intelligent behavior can be extracted from optimization over vast parameter spaces: neural networks with billions of weights, trained by gradient descent against differentiable objectives, produce capabilities no hand-designed architecture reaches. A structurally analogous revolution has been unfolding in physics. Photonic crystals engineer electromagnetic band structure through lattice geometries no monatomic

medium provides; metasurfaces compress wavefront-shaping into subwavelength films through the orchestration of millions of sub-wavelength scatterers; inverse-designed silicon photonics [1–6] achieves mode converters, spectral filters, and freeform functional elements in footprints no hand-designed waveguide can reach; acoustic and elastic metamaterials confer effective negative moduli through sub-unit-cell resonances; topological photonics and iridescent biological structures exploit long-range geometric order to produce behavior no constituent chemistry can. The template is the same as deep learning's: a vast parameter space, namely the geometry of a physical medium, optimized by gradient descent against a differentiable physical objective (scattering amplitudes, information content, coherent-addition structure) produces functional behavior no simple structure can. In this sense, *structural complexity in physics is the analog of representational complexity in deep networks*: both are substrates through which optimization converts raw degrees of freedom into engineered capability. The consequence is that physical hardware optimization is not a peripheral engineering concern but a fundamental question about where intelligence can live.

This question sharpens dramatically in sensing and metrology, at the analog-to-digital boundary where a physical signal becomes a digital record and any information lost there bounds everything any downstream algorithm can recover. The conventional blueprint of the last two decades pairs a generic, task-agnostic transducer front-end with powerful digital reconstruction, and so treats the physical hardware as a passive conduit and asks the digital stages to rescue whatever structure the geometry happens to impose. Computational inverse design already inverts this blueprint on the hardware side, routinely saturating the electromagnetic performance bounds of the underlying scattering problem [3, 5, 7, 8], and end-to-end optical co-design extends it by tying the geometry to a downstream reconstruction network, closing the loop from physics to inference in a single gradient [9–18]. What remains missing is the policy dimension: the recognition that the geometry must co-design with *a rule for what to measure next*, i.e., a multi-step adaptive measurement policy that reacts to intermediate data. When such a policy is installed, part of the intelligence-extraction work migrates upstream into the analog domain before digitization, and the co-optimized geometry becomes a participant in the inference itself rather than a conduit to it. This paper supplies that missing dimension.

A mature literature on adaptive sensing and sequential Bayesian experimental design constructs formally-optimal multi-step measurement policies for partially-observable Markov decision processes (POMDPs) [19–31], but treats the physical hardware as immutable. The hardware half (inverse design, end-to-end optics) and the policy half (adaptive-sensing POMDP theory) are thus designed separately, each community optimizing its own half against the other's default: a textbook setup for a joint optimum that nobody reaches. Several near-relatives live in the intersection without closing it: cognitive radar adapts the waveform on a fixed antenna [32–35], Kitaev-style quantum phase estimation adapts the pulse sequence on a fixed circuit [36–42], programmable-metasurface imaging adapts the reconstruction on a fixed scattering medium [43, 44], and variational quantum sensing parametrizes the probe through a trainable neural ansatz rather than a geometric topology [45, 46]. None of these treats continuous physical geometry and a multi-step Bellman-optimal measurement policy as co-equal design variables in a single optimization.

Closing this gap requires a framework that (a) is cleanly defined across discrete and continuous state spaces, (b) admits a tractable outer gradient with respect to the physical geometry, and (c) offers a hierarchy of relaxations scaling from exact dynamic programming on small POMDPs to high-dimensional continuous hardware inverse design. We supply all three, and call the procedure *joint dynamic programming*, or *joint-DP*: a single optimization that treats the continuous hardware $\boldsymbol{c}$ and the Bellman-optimal adaptive policy π as co-equal decision variables, with π re-solved by dynamic programming at each outer iterate in $\boldsymbol{c}$. To our knowledge, this is the first formulation in which continuous physical hardware and a multi-step Bellman-optimal adaptive policy are co-designed as a single optimization problem.

Joint-DP improves on the two natural baselines that bracket the problem, on benchmarks where exact comparison is tractable. The adaptive-sensing POMDP literature runs the Bellman dynamic-programming (DP) recursion (the standard backward-in-time value-iteration procedure that solves for an optimal measurement policy at fixed hardware) joint-DP strictly generalizes this by co-optimizing $\boldsymbol{c}$ under the adaptive policy, yielding a 123× deployed-MSE improvement on the photonic metasensor of Case Study C over a non-co-designed baseline. The OED and physics-of-bounds literatures select $\boldsymbol{c}$ to maximize a prior-averaged, fixed-schedule information functional (mutual information, expected-Fisher log-determinant, Expected Value of Perfect Information) and attach an adaptive policy afterwards; on the radar POMDP of Case Study A the information-functional-maximizing $\boldsymbol{c}$ loses a factor of 2.8× in attainable adaptive value to the joint-DP optimum, and on the superconducting-qubit flux sensor of Case Study B the joint Bayesian Cramér–Rao fixed-schedule baseline loses a factor of 11.3× in deployed Bayesian MSE ($z = +132\sigma$). The performance gaps are first-order in engineering terms.

1.1 Contributions

1. **Unified formulation.** A family of pointwise scalar rewards $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ (mutual-information-aligned Φ_{MI} [27], Bayesian-MSE-aligned Φ_{Var}, and Fisher-log-det $\Phi_{\log J}$ [47, 48]) fit the same joint-$(\boldsymbol{c}, \pi)$ Bellman machinery. The pointwise dependence on the unknown state $\boldsymbol{x}$ is the native reward form of POMDP theory, but it is rarely foregrounded in the adaptive-sensing and Bayesian-OED communities, which typically work directly with prior-averaged belief functionals; foregrounding it is what unifies three traditionally separate reward families under one umbrella and what makes the oracle / fixed / adaptive decomposition of §3 well-defined. Choosing Φ is a modeling commitment set by the problem's deployment metric.

2. **Structural insight.** The envelope theorem [49] (the optimality condition that lets one differentiate a maximum-over-π expression by holding π fixed at its optimum) eliminates the policy Jacobian $\partial\pi^\star/\partial\boldsymbol{c}$ from the outer geometry gradient, making joint optimization tractable even when the inner Bellman solver is approximate.

3. **Quantitative findings.** On benchmarks where exact comparison is tractable, joint-DP delivers performance gaps of 2.8× against classical information-bound-guided geometry selection (radar, Case Study A); 11.3× at $z = +132\sigma$ in deployed Bayesian MSE against the joint Bayesian Cramér–Rao baseline [50, 51] (scqubit, Case Study B); and 123× in deployed MSE against a non-co-designed baseline (photonic, Case Study C).

4. **Portability across scales.** A relaxation cascade (exact DP, certainty-equivalent, myopic one-step, Fisher-trace plus EKF) carries the framework from small POMDPs to a 90,000-pixel photonic metasensor, with each rung's failure mode named.

1.2 Relation to prior work

The components of our framework are individually well-established: exact Bellman DP for POMDPs [20, 21, 52]; mutual information as an experimental-design reward (Lindley [27]); end-to-end optical co-design with learned reconstruction, including information-reward end-to-end imaging by Waller and collaborators [53–56]; the envelope/Danskin theorem [49, 57]; Expected Value of Perfect Information as a classical decision-theoretic quantity [58, 59]. The joint optimization of *continuous* physical hardware with a *multi-step Bellman-optimal* adaptive policy is not standard in any of these: end-to-end optics uses one-step reconstructions, adaptive sensing fixes the hardware, and the EVPI/IG line does not treat continuous physical geometry as a first-class design variable. Each ingredient is classical; the novelty is their synthesis into a

single joint optimization over continuous physical hardware and a multi-step Bellman-optimal adaptive policy, with the envelope theorem as the enabling identity on the outer gradient.

Our framework sits downstream of, and is complementary to, the recent program on fundamental limits to light–matter interaction via convex-duality and T-matrix methods [7, 8, 60–71]: those bounds certify whether a target is achievable by *any* passive medium; we select the $\boldsymbol{c}$ that realizes it along with the policy that consumes it. Likewise for the quantum Fisher information program on wave and photon scattering [72–78]: our $\Phi_{\log J}$ choice in the Gaussian-posterior regime is the Bayesian analog of the maximum-Fisher-information-state object; outside that regime it shares the stage with alternative scalar rewards chosen by the problem's deployment metric.

1.3 Organization

The paper builds up from formulation to evidence. Section 2 states the joint-$(\boldsymbol{c}, \pi)$ optimization and develops the family of pointwise scalar rewards Φ. Section 3 introduces the value decomposition and the ignorance gap. Section 4 derives the envelope-theorem outer gradient; Section 5 catalogs the relaxation hierarchy. Sections 6, 7, and 8 are the three case studies (discrete POMDP; continuous-state qubit sensor; high-dimensional photonic metasensor). Sections 9–10 discuss limitations and conclude.

2 Problem setup and the scalar-reward family

2.1 Sensor model

A *sensor* in this paper is a tuple $(\boldsymbol{c}, \pi)$ where $\boldsymbol{c} \in \mathcal{C} \subseteq \mathbb{R}^{d_c}$ is a continuous *physical geometry* fixed at manufacturing time (the flux-bias and readout-line mutual inductances of a SQUID loop, the permittivity distribution of an inverse-designed metasurface, the elements of a phased-array antenna), and $\pi : \mathcal{B} \to \mathcal{S}$ is a *measurement policy* that maps the current belief $\boldsymbol{b} \in \mathcal{B}$ over the unknown state $\boldsymbol{x} \in \mathcal{X}$ to an action $\boldsymbol{s} \in \mathcal{S}$. At each step $k \in \{1, \ldots, K\}$ the policy emits an action $\boldsymbol{s}_k = \pi(\boldsymbol{b}_k)$, whose execution produces a *measurement* $\boldsymbol{y}_k \in \mathcal{Y}$ drawn from the likelihood

$$\boldsymbol{y}_k \;\sim\; p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k;\, \boldsymbol{c}), \tag{1}$$

where $\mathcal{Y}$ is the measurement space specified by the physical readout (discrete detector counts in a photon-starved regime, photon clicks in a Ramsey sequence, real-valued voltages from a homodyne detector, etc.) and $\boldsymbol{c}$ parametrizes the likelihood through its physical forward model (Maxwell's equations, circuit Hamiltonian dynamics, Schrödinger evolution with pulse shaping, etc.). For non-adaptive ("fixed-schedule") policies π is a constant function of the belief; for adaptive policies π depends on the history $\mathcal{F}_{k-1} = \{\boldsymbol{s}_{1:k-1}, \boldsymbol{y}_{1:k-1}\}$ through the posterior $\boldsymbol{b}_k = p(\boldsymbol{x} \mid \mathcal{F}_{k-1};\, \boldsymbol{c})$. We write K for the *horizon* — the total number of measurement steps the operator is budgeted before an estimate of $\boldsymbol{x}$ is reported, which may be limited by coherence time in a quantum sensor, by integration-time budget in a radar search, or by throughput requirements in an imaging pipeline — $p_0(\boldsymbol{x})$ for the prior, and reserve $\pi^\star(\boldsymbol{c})$ for the Bellman-optimal policy at fixed $\boldsymbol{c}$. Inner expectations are over observations $\boldsymbol{y}$ with $\boldsymbol{x}, \boldsymbol{c}$ held fixed; outer expectations are over the prior p_0.

2.2 Joint optimization statement

A *joint co-design objective* is any scalar functional of the hardware-policy pair,

$$V(\boldsymbol{c}, \pi) \;=\; \mathbb{E}_{\boldsymbol{x}\sim p_0}\, \mathbb{E}_{\boldsymbol{y}_{1:K}|\boldsymbol{x},\pi,\boldsymbol{c}}\big[\, \Phi(\boldsymbol{x}, \boldsymbol{s}^{\pi}_{1:K}, \boldsymbol{c})\,\big], \tag{2}$$

where $\Phi : \mathcal{X} \times \mathcal{S}^K \times \mathcal{C} \to \mathbb{R}$ is a user-specified *pointwise reward*: "pointwise" in the sense that Φ is evaluated at each realization of the true state $\boldsymbol{x}$, action sequence $\boldsymbol{s}_{1:K}$, and hardware $\boldsymbol{c}$

separately, before any averaging over the prior. Aggregate objectives (Bayesian mean-squared error, mutual information, log-determinant of an expected Fisher matrix) are recovered by taking the expectations in Equation (2). We correspondingly use "pointwise MSE" for the state-conditional mean-squared error $\mathbb{E}_{\boldsymbol{y}|\boldsymbol{x}}[(\hat{\boldsymbol{x}} - \boldsymbol{x})^2]$, which is an $\boldsymbol{x}$-dependent scalar, as opposed to the aggregate Bayesian MSE which is its prior average. The joint hardware–policy optimization solves

$$(\boldsymbol{c}^\star, \pi^\star) \;=\; \underset{\boldsymbol{c}\in\mathcal{C},\, \pi\in\Pi}{\arg\max}\; V(\boldsymbol{c}, \pi). \tag{3}$$

Equation (3) is the sole formal statement in the paper; what varies across applications is only the choice of Φ. Two degenerate cases recover classical programs: $\Pi = \{\text{constant policies}\}$ with Φ minus a reconstruction loss recovers **end-to-end optics** [9, 10]; $\mathcal{C} = \{\boldsymbol{c}_0\}$ (fixed hardware) with any Φ recovers **adaptive sensor management** [19, 22]. Their pairing is the content of this paper.

For the posterior-mean estimator on continuous $\boldsymbol{x}$, the tower-law identity $\mathbb{E}[(\hat{\boldsymbol{x}}_{\mathrm{pm}} - \boldsymbol{x})^2] = \mathbb{E}[\mathrm{Var}(\boldsymbol{x} \mid \boldsymbol{b}_K)]$ makes the Bayesian MSE a special case of $V(\boldsymbol{c}, \pi)$ with $\Phi = -\mathrm{Var}(\boldsymbol{x} \mid \boldsymbol{b}_K)$. Section 2.3 develops this and two alternative Φ choices used by the other case studies.

Two-level structure (joint-DP). When Π is the full space of adaptive policies, the inner $\arg\max_\pi$ in Equation (3) is solved by the Bellman recursion on the belief state $\boldsymbol{b}_k$ (derivation from Equation (2) in Appendix A),

$$V_K(\boldsymbol{b};\, \boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{x}|\boldsymbol{b}}[\,\Phi_0(\boldsymbol{x}, \boldsymbol{b};\, \boldsymbol{c})\,], \tag{4}$$

$$V_k(\boldsymbol{b};\, \boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_k\in\mathcal{S}}\; \mathbb{E}_{\boldsymbol{y}_k|\boldsymbol{b},\boldsymbol{s}_k,\boldsymbol{c}}\big[\,V_{k+1}\big(T(\boldsymbol{b}, \boldsymbol{s}_k, \boldsymbol{y}_k;\, \boldsymbol{c});\, \boldsymbol{c}\big)\,\big], \quad k = K-1, \ldots, 0, \tag{5}$$

where $T(\boldsymbol{b}, \boldsymbol{s}, \boldsymbol{y};\, \boldsymbol{c})$ is the Bayesian belief-update operator,

$$T(\boldsymbol{b}, \boldsymbol{s}, \boldsymbol{y};\, \boldsymbol{c})(\boldsymbol{x}) \;=\; \frac{p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{s};\, \boldsymbol{c})\, \boldsymbol{b}(\boldsymbol{x})}{\int_{\mathcal{X}} p(\boldsymbol{y} \mid \boldsymbol{x}', \boldsymbol{s};\, \boldsymbol{c})\, \boldsymbol{b}(\boldsymbol{x}')\, d\boldsymbol{x}'}, \tag{6}$$

the posterior over $\boldsymbol{x}$ obtained from prior $\boldsymbol{b}$ by Bayes' rule given one measurement $\boldsymbol{y}$ taken at action $\boldsymbol{s}$ under hardware $\boldsymbol{c}$ (the integral in the denominator is a sum for discrete $\boldsymbol{x}$). $\Phi_0(\boldsymbol{x}, \boldsymbol{b}_K;\, \boldsymbol{c})$ is the terminal pointwise reward written as a function of the terminal belief; for $\Phi_{\log J}$ the belief is augmented by the accumulated Fisher-information matrix, and Φ_0 is read off the augmentation. The inner optimal policy at each step is $\pi_k^\star(\boldsymbol{b};\, \boldsymbol{c}) = \arg\max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k}[V_{k+1}(T(\boldsymbol{b}, \boldsymbol{s}_k, \boldsymbol{y}_k;\, \boldsymbol{c});\, \boldsymbol{c})]$, and the joint hardware-policy optimum is the outer maximum of the inner DP value at the initial belief $\boldsymbol{b}_0 = p_0$,

$$\boxed{\boldsymbol{c}^\star \;=\; \underset{\boldsymbol{c}\in\mathcal{C}}{\arg\max}\; V^\star(\boldsymbol{c}), \qquad V^\star(\boldsymbol{c}) \;\coloneqq\; V_0(\boldsymbol{b}_0;\, \boldsymbol{c}).} \tag{7}$$

We call Equations (4)–(7) the *joint-DP* structure: an outer continuous optimization over the physical geometry $\boldsymbol{c}$ wrapped around an inner Bellman recursion that produces the optimal adaptive policy $\pi^\star(\boldsymbol{c})$ at each candidate $\boldsymbol{c}$. Section 4 gives an envelope-theorem outer gradient that decouples the $\boldsymbol{c}$-gradient from the inner Bellman solve; Section 5 describes a cascade of relaxations for the inner recursion when exact DP is infeasible.

2.3 The scalar-reward family Φ

Three canonical choices for Φ emerge naturally from three different bodies of theory. In each case Φ is a deterministic function of $(\boldsymbol{x}, \boldsymbol{s}_{1:K}, \boldsymbol{c})$; any expectation over the measurement trajectory $\boldsymbol{y}_{1:K}$ that appears in the definition is baked in, so that the outer objective $V(\boldsymbol{c}, \pi) = \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}}[\Phi]$ of Equation (2) is a single expectation over the prior and the data.

Mutual-information-aligned reward $\Phi_{\mathbf{MI}}$. Following the Shannon–Lindley line [27, 79] through Chaloner–Verdinelli [31], Sebastiani–Wynn [80], and the adaptive sensor-management literature [22, 23],

$$\Phi_{\mathrm{MI}}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{y}|\boldsymbol{x},\boldsymbol{s},\boldsymbol{c}}\big[\ln|\mathrm{supp}(\boldsymbol{b}_K)| - H(\boldsymbol{b}_K)\big], \tag{8}$$

the expected negative entropy of the terminal belief $\boldsymbol{b}_K$ at state $\boldsymbol{x}$. Averaging over the prior gives $\mathbb{E}_{\boldsymbol{x}}[\Phi_{\mathrm{MI}}] = I(\boldsymbol{x}; \boldsymbol{y}_{1:K} \mid \boldsymbol{s}, \boldsymbol{c})$, the mutual information. This is the reward of choice for discrete-state problems where MSE is undefined or dominated by 0/1 classification loss [24], and for any problem in the large-K Bernstein–von Mises regime [81, 82] (the asymptotic limit in which the posterior converges to a Gaussian centered near the true parameter, making MI monotone in MSE). The photonic and quantum-sensing communities have developed extensive machinery for *upper bounds* on achievable mutual information given physical constraints (reciprocity, passivity, energy); our framework consumes those bounds as constraints on the feasible set $\{\Phi_{\mathrm{MI}}(\cdot, \cdot, \boldsymbol{c}) : \boldsymbol{c} \in \mathcal{C}\}$ rather than re-deriving them.

Bayesian-MSE-aligned reward $\Phi_{\mathbf{Var}}$. Inheriting from the Bayesian Cramér–Rao tradition [50] and the posterior-variance-minimization line of Bayesian optimal experimental design [31, 83], the tower law of variance (the identity that the Bayes-averaged squared error of the posterior mean equals the Bayes-averaged posterior variance)

$$\mathbb{E}_{\boldsymbol{x},\boldsymbol{y}_{1:K}}\big[(\hat{\boldsymbol{x}}_{\mathrm{pm}} - \boldsymbol{x})^2\big] \;=\; \mathbb{E}_{\boldsymbol{y}_{1:K}}[\mathrm{Var}(\boldsymbol{x} \mid \boldsymbol{y}_{1:K})] \tag{9}$$

(with $\hat{\boldsymbol{x}}_{\mathrm{pm}}$ the posterior mean) motivates

$$\Phi_{\mathrm{Var}}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{y}|\boldsymbol{x},\boldsymbol{s},\boldsymbol{c}}\big[-\mathrm{Var}(\boldsymbol{x} \mid \boldsymbol{b}_K)\big], \tag{10}$$

for which the outer objective $V(\boldsymbol{c}, \pi)$ equals $-1\times$ the deployed Bayesian MSE exactly. The joint-DP optimum of this reward is literally the *MSE-optimal* adaptive policy at fixed $\boldsymbol{c}$, without any asymptotic approximation. For continuous-state, finite-K problems with bounded-support priors, where the Bernstein–von Mises regime is not reached, Φ_{Var} strictly dominates Φ_{MI} as an MSE anchor. The tower-law identity is textbook; its use as a joint-DP terminal reward in the present hardware-policy setting is new.

Fisher-information-aligned reward $\Phi_{\log J}$. Inheriting from the classical Fisher-information-matrix tradition [47, 48, 84] and its Bayesian extension [50, 51], and applicable to continuous $\boldsymbol{x}$ with a Gaussian-posterior surrogate,

$$\Phi_{\log J}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) \;=\; \tfrac{1}{2}\log\det J_N(\boldsymbol{x}, \boldsymbol{s}_{1:K}, \boldsymbol{c}), \tag{11}$$

where J_N is the accumulated Fisher information matrix at true $\boldsymbol{x}$. The outer expectation is the log-determinant of the Bayesian Fisher information matrix [31, 85], with J_N^{-1} the Bayesian CRB. In the Gaussian-posterior regime $\Phi_{\log J}$ agrees with Φ_{Var} up to log-transforms. Outside that regime (biased estimators, bounded priors, small K), $\Phi_{\log J}$ retains an asymptotic CRB interpretation while losing its direct link to finite-sample MSE. It is the natural reward for high-dimensional continuous problems where the full Bellman DP is intractable and the Gaussian surrogate is the cheapest defensible relaxation. The quantum-Fisher-information program on wave and photon scattering [72–74, 76] and the photonic information-bound program on scattering-matrix extremization [7, 8, 60–62] establish upper bounds on $\det J_N$ over admissible $\boldsymbol{c}$; they tell us what $\sup_{\boldsymbol{c}} \det J_N$ is, and our framework tells us which $\boldsymbol{c}$ and which policy realize it jointly.

Table 1: Three natural pointwise scalar rewards, their operational semantics, and their alignment with finite-K MSE.

reward	terminal expression	community	tied to pointwise MSE?
Φ_{MI}	$\ln \lvert\mathrm{supp}(\boldsymbol{b}_K)\rvert - H(\boldsymbol{b}_K)$	sensor management, sOED	asymptotic only
Φ_{Var}	$-\mathrm{Var}(\boldsymbol{x} \mid \boldsymbol{b}_K)$	Bayesian estimation	**directly, by tower law**
$\Phi_{\log J}$	$\frac{1}{2}\log\det J_N(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$	Fisher info, PCRB	asymptotic only

2.4 Comparison and operational guidance

The three rewards agree asymptotically under Bernstein–von Mises but diverge at finite horizons, and the choice of Φ matters operationally for the co-design optimum at modest horizon on a bounded prior. We accordingly present the paper's machinery as reward-agnostic: the user picks Φ based on the problem's state space, horizon, and deployment metric, and the envelope-theorem outer gradient of Section 4 derives identically from any of them.

3 Value decomposition and the ignorance gap

For any choice of Φ, the outer objective $V(\boldsymbol{c}, \pi)$ admits a four-functional decomposition that is useful for both intuition and diagnostics. The decomposition is standard in sequential Bayesian experimental design [28, 29] and was used in the radar sensor-management literature under the name "oracle value" of perfect information [22, 86]; we restate it in the Φ-agnostic form used in the rest of the paper.

3.1 Four functionals

$$V_{\mathrm{oracle}}(\boldsymbol{x}, \boldsymbol{c}) = \max_{\boldsymbol{s}_{1:K}} \Phi(\boldsymbol{x}, \boldsymbol{s}_{1:K}, \boldsymbol{c}), \tag{12}$$

$$V_{\mathrm{fixed}}(\boldsymbol{c}) = \max_{\boldsymbol{s}_{1:K}} \mathbb{E}_{\boldsymbol{x}}\big[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:K}, \boldsymbol{c})\big], \tag{13}$$

$$V_{\mathrm{adaptive}}(\boldsymbol{c}) = \max_{\pi \in \Pi} \mathbb{E}_{\boldsymbol{x}}\mathbb{E}_{\boldsymbol{y}|\boldsymbol{x},\pi,\boldsymbol{c}}\big[\Phi(\boldsymbol{x}, \boldsymbol{s}^{\pi}_{1:K}, \boldsymbol{c})\big], \tag{14}$$

$$\mathbb{E}[\mathrm{IG}](\boldsymbol{c}) = \mathbb{E}_{\boldsymbol{x}}[V_{\mathrm{oracle}}(\boldsymbol{x}, \boldsymbol{c})] - V_{\mathrm{fixed}}(\boldsymbol{c}). \tag{15}$$

The functional $V_{\mathrm{oracle}}(\boldsymbol{x}, \boldsymbol{c})$ is the value attained by a *clairvoyant* sensor that knows the true $\boldsymbol{x}$ and picks the best fixed schedule for it; its prior average is an oracle upper bound. $V_{\mathrm{fixed}}(\boldsymbol{c})$ is the best non-adaptive schedule. $V_{\mathrm{adaptive}}(\boldsymbol{c})$ is the best adaptive policy, which is the quantity the user ultimately deploys. $\mathbb{E}[\mathrm{IG}](\boldsymbol{c})$, the *expected ignorance gap* or *expected value of perfect information* (EVPI) [58, 59], is a well-defined scalar functional of $\boldsymbol{c}$ that measures how much adaptive slack exists at that geometry.

3.2 Inequalities and the adaptive gain

The three values are ordered at every $\boldsymbol{c}$,

$$V_{\mathrm{fixed}}(\boldsymbol{c}) \;\le\; V_{\mathrm{adaptive}}(\boldsymbol{c}) \;\le\; \mathbb{E}_{\boldsymbol{x}}[V_{\mathrm{oracle}}(\boldsymbol{x}, \boldsymbol{c})], \tag{16}$$

because fixed schedules are a subset of adaptive policies (the left inequality) and no policy can beat the clairvoyant oracle (the right inequality). The *adaptive gain* at hardware $\boldsymbol{c}$ is the nonnegative difference $V_{\mathrm{adaptive}}(\boldsymbol{c}) - V_{\mathrm{fixed}}(\boldsymbol{c})$, and subtracting V_{fixed} from Equation (16) bounds it above by the EVPI,

$$0 \;\le\; V_{\mathrm{adaptive}}(\boldsymbol{c}) - V_{\mathrm{fixed}}(\boldsymbol{c}) \;\le\; \mathbb{E}[\mathrm{IG}](\boldsymbol{c}). \tag{17}$$

Equation (17) is what makes the EVPI a candidate design criterion: if the upper bound were tight, the geometry that maximizes $\mathbb{E}[\mathrm{IG}]$ would coincide with the one that maximizes the adaptive gain. Whether that hope survives scrutiny is the topic of §3.4.

3.3 Why joint hardware–policy co-design is necessary

The quantity the user deploys is $V_{\text{adaptive}}(\boldsymbol{c})$, so the question is which hardware maximizes that deployed value. Two facts make it a function of $\boldsymbol{c}$ that cannot be maximized piecewise.

Fact 1: adaptation extracts real value at most geometries. A fixed schedule commits all K measurements in advance, before any data has arrived. An adaptive policy sees each outcome as it happens and re-routes subsequent measurements accordingly: effort that would have been wasted confirming something the data has already settled gets redirected toward parts of the state space that remain ambiguous. In an imaging problem this is the difference between a fixed raster scan and one that zooms in on the suspected region; in a qubit-flux sensor it is the difference between a fixed Ramsey delay ladder and one that shortens τ when the posterior is broad (to avoid phase-wrapping ambiguity) and lengthens τ once the flux is roughly localized. The adaptive gain of Equation (17) is therefore typically strictly positive and often sizable.

Fact 2: the adaptive gain depends nontrivially on $\boldsymbol{c}$. A hardware configuration that maximizes V_{fixed} is not in general the hardware that maximizes V_{adaptive}: a geometry good for blind schedules is good because its expected measurements are individually informative on average, and that property tends to leave little room for outcome-dependent re-routing. Conversely, a geometry with very large oracle value (the right-hand side of Equation (16)) may put so much of that value out of reach of any policy that only sees past data that the realizable adaptive value is actually small. The geometry that deploys well under an adaptive policy is an *interior* geometry, one that neither is over-tuned to blind schedules nor relies on clairvoyance; this optimum is not located by examining V_{fixed} or V_{oracle} separately.

Consequence: the two layers cannot be solved in sequence. One cannot pick $\boldsymbol{c}$ first (by any single-functional criterion) and then pick π, because the criterion for a good $\boldsymbol{c}$ depends on which π will follow. Joint optimization of $(\boldsymbol{c}, \pi)$ is necessary; the rest of the paper is about making it tractable. Section 4 gives the envelope-theorem identity that decouples the *gradient computation* from the inner-policy solve even when the two quantities themselves do not decouple, so that an outer-loop optimizer can climb $V_{\text{adaptive}}(\boldsymbol{c})$ with a single Bellman solve per iteration.

3.4 Why EVPI does not identify the right geometry

Before committing to the joint solve, it is worth asking whether the EVPI can stand in. Since V_{oracle} and V_{fixed} are computable by direct schedule enumeration (no Bellman recursion, no belief tree), $\mathbb{E}[\mathrm{IG}](\boldsymbol{c})$ is a cheap scalar functional of $\boldsymbol{c}$, unlike $V_{\text{adaptive}}(\boldsymbol{c})$ which requires the full backward-in-time value iteration. One might hope that maximizing the easy functional is enough.

It is not: the upper bound in Equation (17) is typically slack, and its slackness pattern is non-monotone in $\boldsymbol{c}$, so

$$\arg\max_{\boldsymbol{c}} \mathbb{E}[\mathrm{IG}](\boldsymbol{c}) \quad \neq \quad \arg\max_{\boldsymbol{c}} V_{\text{adaptive}}(\boldsymbol{c}). \tag{18}$$

The geometry with the largest in-principle adaptive gain, as measured by EVPI, is typically a geometry where most of that gain is unreachable in practice. Extreme geometries (a very narrow

beam, a near-isotropic beam) produce high EVPI because the oracle's advantage over any blind fixed schedule is enormous, but the adaptive policy, which does not know $\boldsymbol{x}$, cannot replicate the oracle's strategy and captures only a small fraction of the nominal headroom. Interior geometries have a modest EVPI ceiling but let the adaptive policy capture a larger fraction of it, which is what ultimately matters. On the radar benchmark of Section 6, $\boldsymbol{c}^{\mathrm{EVPI}}$ loses a factor of 2.8× in attainable V_{adaptive} to the joint-DP optimum; the joint solve is unavoidable.

3.5 EVPI's remaining roles

Although EVPI cannot replace the joint solve as a design criterion, it retains three useful roles:

1. **A pre-screen when the full joint DP is prohibitive but schedule enumeration is tractable.** $\mathbb{E}[\mathrm{IG}](\boldsymbol{c})$ depends only on V_{oracle} and V_{fixed}; its cheapness is inherited from the cheapness of those enumerations (small discrete action sets or low-dimensional continuous actions on a reasonable grid), and evaporates when schedule enumeration itself becomes the bottleneck.
2. **Post-hoc interpretation.** After solving the joint problem, $\mathbb{E}[\mathrm{IG}](\boldsymbol{c}^\star)$ quantifies the headroom between the attained adaptive value and the oracle ceiling, indicating where further gains might come from (longer horizon, richer action set).
3. **Diagnostic lens on the slackness pattern.** When $\arg\max_{\boldsymbol{c}} \mathbb{E}[\mathrm{IG}]$ and $\arg\max_{\boldsymbol{c}} V_{\mathrm{adaptive}}$ disagree, the disagreement structure exposes the physics: extreme geometries leave the most adaptive headroom on the table, and interior geometries capture more of a smaller ceiling. Section 6 develops this in detail.

4 Envelope-theorem gradient for the outer loop

4.1 Statement

Let $\pi^\star(\boldsymbol{c}) = \arg\max_{\pi\in\Pi} V(\boldsymbol{c},\pi)$ be the optimal policy at hardware $\boldsymbol{c}$. The joint-optimal geometry solves

$$\boldsymbol{c}^\star = \arg\max_{\boldsymbol{c}} V^\star(\boldsymbol{c}), \qquad V^\star(\boldsymbol{c}) \coloneqq V(\boldsymbol{c},\pi^\star(\boldsymbol{c})). \tag{19}$$

A naive outer gradient requires differentiating through the policy argmax:

$$\frac{dV^\star}{d\boldsymbol{c}} = \left.\frac{\partial V}{\partial \boldsymbol{c}}\right|_{\pi=\pi^\star(\boldsymbol{c})} + \underbrace{\left.\frac{\partial V}{\partial \pi}\right|_{\pi=\pi^\star(\boldsymbol{c})}}_{=0 \text{ at optimum}} \cdot \frac{d\pi^\star}{d\boldsymbol{c}}. \tag{20}$$

The second term vanishes at the optimum by the first-order optimality condition (or, in discrete-action settings, by the envelope/Danskin theorem [49, 57, 87]). Consequently the outer gradient is simply

$$\boxed{\frac{dV^\star}{d\boldsymbol{c}} = \left.\frac{\partial V(\boldsymbol{c},\pi^\star(\boldsymbol{c}))}{\partial \boldsymbol{c}}\right|_{\pi^\star \text{ held fixed}}.} \tag{21}$$

The policy Jacobian $d\pi^\star/d\boldsymbol{c}$ does not appear.

4.2 Unfolding the envelope gradient through the Bellman recursion

Equation (21) gives the envelope gradient in its abstract form. Making it operational requires unfolding through the Bellman recursion of Equations (4)–(5) to identify the concrete paths by which $\boldsymbol{c}$ enters $V^\star$. The result is a backward-recursion formula for $\partial V_k/\partial \boldsymbol{c}$ that mirrors the Bellman recursion for V_k, and that can be computed by reverse-mode automatic differentiation with a single structural step: freeze the argmax.

Define the Q-function

$$Q_k(\boldsymbol{b}, \boldsymbol{s}_k;\, \boldsymbol{c}) \;\coloneqq\; \mathbb{E}_{\boldsymbol{y}_k|\boldsymbol{b},\boldsymbol{s}_k,\boldsymbol{c}}\big[V_{k+1}\big(T(\boldsymbol{b}, \boldsymbol{s}_k, \boldsymbol{y}_k;\, \boldsymbol{c});\, \boldsymbol{c}\big)\big], \tag{22}$$

so that the Bellman recursion reads $V_k(\boldsymbol{b};\, \boldsymbol{c}) = \max_{\boldsymbol{s}_k} Q_k(\boldsymbol{b}, \boldsymbol{s}_k;\, \boldsymbol{c})$ with $\boldsymbol{s}_k^\star(\boldsymbol{b};\, \boldsymbol{c}) \coloneqq \arg\max_{\boldsymbol{s}_k} Q_k$. The total derivative of V_k with respect to $\boldsymbol{c}$ splits into an indirect piece (via $\boldsymbol{s}_k^\star$) and a direct piece:

$$\frac{\partial V_k}{\partial \boldsymbol{c}}(\boldsymbol{b};\, \boldsymbol{c}) \;=\; \underbrace{\frac{\partial Q_k}{\partial \boldsymbol{s}_k}(\boldsymbol{b}, \boldsymbol{s}_k^\star;\, \boldsymbol{c}) \cdot \frac{\partial \boldsymbol{s}_k^\star}{\partial \boldsymbol{c}}(\boldsymbol{b};\, \boldsymbol{c})}_{=0 \text{ at the argmax}} \;+\; \frac{\partial Q_k}{\partial \boldsymbol{c}}(\boldsymbol{b}, \boldsymbol{s}_k^\star;\, \boldsymbol{c}). \tag{23}$$

The first term vanishes at an interior argmax by first-order optimality ($\partial Q_k/\partial \boldsymbol{s}_k = 0$ at $\boldsymbol{s}_k^\star$). What survives is the partial derivative of Q_k holding $\boldsymbol{s}_k^\star$ fixed — the per-step envelope identity

$$\frac{\partial V_k}{\partial \boldsymbol{c}}(\boldsymbol{b};\, \boldsymbol{c}) \;=\; \frac{\partial Q_k}{\partial \boldsymbol{c}}(\boldsymbol{b}, \boldsymbol{s}_k^\star;\, \boldsymbol{c}). \tag{24}$$

Unfolding $\partial Q_k/\partial \boldsymbol{c}$ using the product and chain rules, three contributions enter:

$$\begin{aligned}\frac{\partial V_k}{\partial \boldsymbol{c}}(\boldsymbol{b};\, \boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{y}_k|\boldsymbol{b},\boldsymbol{s}_k^\star,\boldsymbol{c}}\Bigg[& \underbrace{\frac{\partial \log p(\boldsymbol{y}_k \mid \boldsymbol{b}, \boldsymbol{s}_k^\star, \boldsymbol{c})}{\partial \boldsymbol{c}}\, V_{k+1}\big(T(\cdots);\, \boldsymbol{c}\big)}_{\text{(i) score / outcome-likelihood term}} \\ & + \underbrace{\frac{\partial V_{k+1}}{\partial \boldsymbol{b}}\big(T;\, \boldsymbol{c}\big) \cdot \frac{\partial T(\boldsymbol{b}, \boldsymbol{s}_k^\star, \boldsymbol{y}_k;\, \boldsymbol{c})}{\partial \boldsymbol{c}}}_{\text{(ii) belief-update term}} \\ & + \underbrace{\frac{\partial V_{k+1}}{\partial \boldsymbol{c}}\big(T;\, \boldsymbol{c}\big)}_{\text{(iii) recursive term}} \Bigg],\end{aligned} \tag{25}$$

with terminal condition $\partial V_K/\partial \boldsymbol{c} = \partial \Phi_0(\boldsymbol{x}, \boldsymbol{b}_K;\, \boldsymbol{c})/\partial \boldsymbol{c}$ at the terminal belief. The three terms pick up the three ways $\boldsymbol{c}$ enters the Bellman backup once the argmax is frozen: through the outcome distribution (i), through the Bayesian belief update T (ii), and through the recursive dependence of V_{k+1} itself (iii).

Equation (25) mirrors Equation (5): same DP graph, same backward sweep order, same memoization structure. If V_k fits in memory — whether as an exact hash table, a count-tuple dictionary, or a Gaussian-moment parametrization — so does $\partial V_k/\partial \boldsymbol{c}$.

The operational recipe: argmax-freeze, then reverse-mode AD. One does not implement Equation (25) by hand. The workflow is:

1. **Forward pass.** Run backward induction from $k = K$ down to $k = 0$, populating V_k and recording $\boldsymbol{s}_k^\star$ at every reachable belief summary. No AD tape is active.
2. **Gradient pass.** Re-evaluate $V_0(p_0;\, \boldsymbol{c})$ by substituting the recorded $\boldsymbol{s}_k^\star$ in place of $\max_{\boldsymbol{s}_k}$. The resulting expression is argmax-free and, by Equation (21), identically equal to $V^\star(\boldsymbol{c})$.
3. **Reverse-mode AD.** Apply an AD system to the gradient-pass code. Because the expression is composed of closed-form primitives in $\boldsymbol{c}$ (the likelihood, the belief update, the terminal reward) and $\boldsymbol{c}$-free combinators (sums, products, argmax lookups), AD returns $\partial V_0/\partial \boldsymbol{c}$ exactly with no custom rule for the Bellman max.

Freezing the argmax turns the non-smooth max-containing recursion into an ordinary differentiable expression — that is the operational content of the envelope theorem. Without it, AD would try to differentiate through the max and produce subgradient terms that Equation (21) tells us are already zero at the optimum.

Concrete realizations across case studies. In Case Study A (radar, discrete belief), the per-step envelope gradient reduces to a finite sum over reachable beliefs; $\partial \Pr / \partial \boldsymbol{c}$ and $\partial T / \partial \boldsymbol{c}$ have closed forms in the beam-pattern parameter. In Case Study B (qubit, count-tuple state), the only $\boldsymbol{c}$-carrying primitive in the entire DP is the Ramsey likelihood of Equation (28); all of the count-tuple combinatorics, action grid, and memoization are $\boldsymbol{c}$-free, so the reverse-mode AD graph consists of one closed-form function composed through the DP tree via the chain rule. In Case Study C (photonic, Gaussian-posterior surrogate), the envelope theorem specializes to the implicit-function-theorem adjoint over the inner sensor-parameter argmax, and the per-step $\partial Q_k / \partial \boldsymbol{c}$ becomes a closed-form expression in the FDFD S-matrix derivatives; see §8.

4.3 Implications for tractability

Equation (21) is the workhorse of the paper. Four consequences:

1. The outer gradient requires solving the inner problem once (to obtain $\pi^\star(\boldsymbol{c})$) and then differentiating V through the likelihood while holding the policy fixed. This decouples the computational cost of solving the Bellman DP from the cost of computing the outer gradient.
2. When $\pi^\star(\boldsymbol{c})$ is computed only approximately (by deep RL, by an approximate value iteration, by a myopic one-step-lookahead), the envelope theorem still says that at the *approximate* optimum, the approximate policy Jacobian's contribution to the outer gradient is first-order small. The framework accommodates approximate inner solvers without bias.
3. The outer gradient is differentiable in $\boldsymbol{c}$ whenever the likelihood is differentiable in $\boldsymbol{c}$. For photonic problems this is automatic via differentiable electromagnetic simulation [88, 89]; for quantum-circuit problems via differentiable Hamiltonian dynamics [90].
4. The envelope gradient admits a simple Monte-Carlo realization: sample trajectories under $\pi^\star$, differentiate the trajectory log-likelihood times the terminal reward under $\boldsymbol{c}$, average. This is the pathwise REINFORCE estimator [91, 92] specialized to the fixed-policy case.

4.4 Policy iteration

The natural outer-loop algorithm alternates Bellman solves with envelope gradient steps:

Joint $(\boldsymbol{c}, \pi)$ policy iteration:

1. Initialize $\boldsymbol{c}^{(0)}$.
2. For $t = 1, 2, \ldots$ until convergence:
   - Inner: solve $\pi^\star(\boldsymbol{c}^{(t-1)})$ (exact DP, deep RL, myopic, etc.).
   - Outer: $\boldsymbol{c}^{(t)} = \boldsymbol{c}^{(t-1)} + \eta \cdot \frac{\partial V(\boldsymbol{c}^{(t-1)}, \pi^\star(\boldsymbol{c}^{(t-1)}))}{\partial \boldsymbol{c}}$ with step size η.
   - Optionally: box-project $\boldsymbol{c}^{(t)}$ onto the feasible set.

Convergence to a stationary point of $V^\star$ follows from standard policy-iteration theory [93, 94] when the inner solver is exact; the envelope theorem extends this to approximate inner solvers provided the approximation error is $o(\eta)$.

4.5 A note on continuous actions

Nothing in the envelope theorem presupposes discrete actions. For continuous $\boldsymbol{s}$, the Bellman backup's inner maximization is a continuous optimization:

$$V_k(\boldsymbol{b}) = \max_{\boldsymbol{s} \in \mathcal{S}} \mathbb{E}_{\boldsymbol{y}}[V_{k+1}(T(\boldsymbol{b}, \boldsymbol{s}, \boldsymbol{y}))], \tag{26}$$

with the terminal condition V_K given by the inner expression of Φ at the terminal belief (Table 1). The $k = 0$ value under the optimal policy coincides with $V(\boldsymbol{c}, \pi^\star)$ of Equation (2).

Standard tools, differential dynamic programming [95, 96], sample-based tree search [97, 98], local gradient inner optimization, HJB-PDE solutions [99], apply. The one caveat specific to our setting is that the standard count-statistic belief memoization used by discrete-action Bellman DP (Section 6) fails for continuous actions because distinct inner-optimization candidates produce distinct child beliefs. We handle this in Case Study C by using the Gaussian-posterior surrogate, in which the belief state is finite-dimensional and the Bellman backup admits a one-step-lookahead closed form; and in Case Study B by log-spaced discretization of the continuous Ramsey-delay dimension at $J = 10$ grid points.

5 A hierarchy of relaxations

The joint $(\boldsymbol{c}, \pi)$ optimization of Section 2 is exactly solvable only for small discrete POMDPs. When exact Bellman DP is intractable, two distinct philosophies emerge for handling the inner optimization.

Dichotomy at the exact-DP branch point. The dominant approach in modern reinforcement learning is to parametrize the policy π_θ (or the action-value Q_θ) with a neural network and optimize it by policy gradient, Q-learning, or actor–critic [100, 101]. *We do not pursue this branch in the present paper, though we do not rule it out for future work.* The reason is a well-documented pathology: an expressive NN policy can absorb the training signal and compensate for a poor $\boldsymbol{c}$ by investing in policy complexity, leaving $\boldsymbol{c}$ near its initialization. The outer loop converges to a *hardware-light* fixed point that looks good in simulation but relies on a near-trivial $\boldsymbol{c}$ hard to justify as a physical design. Known countermeasures — capacity control on π_θ (bottleneck width, weight decay, dropout), two-timescale updates with $\boldsymbol{c}$ stepped on a faster clock than θ, warm-starts from one of the prescribed inner rules below (rungs 3–4), and auxiliary regularizers penalizing hardware-null geometries — partially mitigate the risk but do not yet admit the clean envelope-theorem outer gradient our framework exploits. We flag the combination of these remedies as a plausible path to extending joint-$(\boldsymbol{c}, \pi)$ to expressive NN policies in future work, and correspondingly exclude NN-parametrized policies from the hierarchy below on a *provisional* rather than principled basis.

The plausible-policy cascade (our path). The alternative, which we pursue, is to *prescribe* π in closed form as a *plausibly Bellman-optimal* rule at the affordable level of fidelity, and to focus the optimization machinery on $\boldsymbol{c}$. Each rule is a closed-form $\arg\max$ over sensor parameters given a belief summary; it picks a principled heuristic that is plausibly close to the Bellman optimum at its feasibility ceiling, then uses the outer loop to move $\boldsymbol{c}$. The hierarchy is a *cascade*: each rung is triggered by the previous becoming intractable.

Because the policy at every rung is prescribed as an explicit function of the current posterior (or one of its coarsenings), this philosophy *requires* an explicit Bayesian filter as the inference primitive — the filter supplies the sufficient statistic the prescribed rule consumes. Explicit inference is therefore a *consequence* of prescribing a plausible policy, not an independent design choice. This is what the cascade shares with exact DP (rung 1) and what sharply distinguishes our branch from the NN branch above, where a learned encoder over the raw observation history replaces the Bayesian filter entirely. What descends the cascade is (i) the policy's proximity to the Bellman optimum — Bellman-optimal at rung 1, clairvoyant-CE at rung 2, myopic one-step at rungs 3–4 — and correspondingly (ii) the coarseness of the belief summary the policy consumes (and the filter maintains) — exact posterior, point estimate, full distribution, Gaussian moments. At no rung is the inference primitive replaced by a learned representation; the Bayesian filter is a first-class object throughout, because the prescribed policy requires it to be. The envelope-theorem gradient of Section 4 relies on this: it applies whenever the inner

rule is a prescribed arg max over a belief-summary sufficient statistic, regardless of how coarsely that summary approximates the true posterior.

Rung 1 — Exact Bellman DP. $\pi_k^\star = \arg\max_{\boldsymbol{s}_k} \mathcal{Q}_k^\star(\boldsymbol{b}_k, \boldsymbol{s}_k)$, the true optimum. Inference: exact belief $\boldsymbol{b}_k$ (tabulated for discrete state; on a K_Φ-point grid for continuous state). Feasibility ceiling: reachable-belief memoization $\sim |\mathcal{S}|^K$ up to $\sim 10^6$ on a single node. When this is violated, move to rung 2.

Rung 2 — Certainty-equivalent control, full horizon. At step k, form a point estimate of the unknown state (posterior mean $\hat{\boldsymbol{x}}_k = \mathbb{E}[\boldsymbol{x} \mid \boldsymbol{b}_k]$ or posterior mode / MAP) and play the schedule the clairvoyant oracle would choose against $\hat{\boldsymbol{x}}_k$. Inference: explicit point estimate from the filter. Plausibly optimal when the posterior concentrates rapidly and the oracle map is Lipschitz in $\boldsymbol{x}$; known failure mode: under-exploration when $\hat{\boldsymbol{x}}_k$ is wrong. Feasibility ceiling: one oracle solve per decision step (itself an N-step enumeration or continuous optimization). When the per-step oracle solve is too expensive, move to rung 3.

Rung 3 — Myopic one-step information criterion. Collapse the look-ahead to one step: $\boldsymbol{s}_k = \arg\max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k|\boldsymbol{b}_k,\boldsymbol{s}_k}\big[\Phi(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})\big]$, i.e., maximize the single-step expected information (or reward) gain conditional on the full current belief. Inference: full belief $\boldsymbol{b}_k$ maintained by exact Bayesian update. Plausibly optimal by submodularity-based near-optimality for information-theoretic rewards [102]; failure mode: myopia proper, i.e., no credit for information that pays off two or more steps out. Feasibility ceiling: one arg max over $\boldsymbol{s}_k$ plus a full-belief expectation per step. When the full-belief expectation is itself too expensive (high-dimensional continuous $\boldsymbol{x}$, for example), move to rung 4.

Rung 4 — Fisher-trace surrogate with EKF. Replace the information criterion by the Fisher-information trace evaluated at the posterior mean, and replace the exact Bayesian update by an extended Kalman filter: $\boldsymbol{s}_k = \arg\max_{\boldsymbol{s}_k} \operatorname{tr} J(\bar{\boldsymbol{x}}_k, \boldsymbol{s}_k, \boldsymbol{c})$ with $\boldsymbol{b}_k \approx \mathcal{N}(\bar{\boldsymbol{x}}_k, \Sigma_k)$ propagated by EKF. Inference: Gaussian moments $(\bar{\boldsymbol{x}}_k, \Sigma_k)$. This is the cheapest rung still compatible with continuous $\boldsymbol{c}$, continuous $\boldsymbol{x}$, and arbitrary state dimension. Failure modes: Gaussian approximation breaks near prior boundaries or under multimodal posteriors; EKF linearization fails when the forward model is strongly non-linear in $\boldsymbol{x}$ over the posterior support.

rung	inner policy rule	inference primitive	triggered when rung above is intractable
1	exact Bellman arg max	exact belief (grid / tabulated)	—
2	certainty-equivalent oracle at $\hat{\boldsymbol{x}}_k$	point estimate $\hat{\boldsymbol{x}}_k$	memoization $\lvert\mathcal{S}\rvert^K \gtrsim 10^6$
3	myopic $\arg\max_{\boldsymbol{s}_k} \mathbb{E}[\Phi \mid \boldsymbol{b}_k]$	full belief $\boldsymbol{b}_k$	per-step oracle too expensive
4	$\operatorname{tr} J(\bar{\boldsymbol{x}}_k, \boldsymbol{s}_k, \boldsymbol{c})$ + EKF	Gaussian $(\bar{\boldsymbol{x}}_k, \Sigma_k)$	full-belief expectation too expensive

Across all four rungs, the envelope theorem applies cleanly to the outer $\boldsymbol{c}$-gradient at the inner arg max, because the inner rule is a prescribed arg max over a belief-summary maintained by an explicit Bayesian filter. The case studies that follow operate at rungs 1, 1, and 4 respectively: Case Study A (radar) at rung 1 with discrete state; Case Study B (scqubit) at rung 1 with a grid-discretized continuous state; Case Study C (photonic metasensor) at rung 4 with $\sim 10^5$ continuous hardware parameters. No single rung uniformly dominates. Each names its

feasibility ceiling and failure mode explicitly so that a practitioner can descend the cascade deliberately, stopping at the first rung that is tractable for the problem in front of them.

An invitation to a broader research program. Once exact Bellman DP is relaxed, the inner policy becomes an approximation, and the full zoo of reinforcement-learning and prescribed-policy frameworks opens up — policy gradient, actor–critic, off-policy Q-learning, POMCP, information-directed sampling, distributional RL, Thompson sampling, model-based planning with learned dynamics, and many more — each with its own strengths, failure modes, and regimes of applicability. To our knowledge, this paper is the first to embed the optimization of continuous physical hardware $\boldsymbol{c}$ into this zoo, via the envelope-theorem outer gradient plus a prescribed, plausibly-Bellman-optimal inner rule. Correspondingly, each existing RL or prescribed-policy framework invites a $\boldsymbol{c}$-aware revision — the "what does this framework look like when the hardware is a co-equal decision variable?" question — whose answer will generally require new theory (ensuring the $\boldsymbol{c}$-gradient is well-defined and free of the hardware-light pathology), new algorithms (coordinating $\boldsymbol{c}$-updates with the inner policy on compatible timescales, integrating regularizers that penalize hardware-null geometries), and new numerics (quantifying how much each framework gains or loses when $\boldsymbol{c}$ is co-optimized rather than fixed). We view this as a new research area opened by the present formulation rather than exhausted by it; the four rungs above are the first systematic traversal, and many more remain.

A deeper stake: intelligence migrating from digital to physical. Information loss at the analog-to-digital boundary upper-bounds any downstream reconstruction; joint $(\boldsymbol{c}, \pi)$ optimization responds by shaping the physical geometry to participate in the inference rather than merely transduce for it. The relaxation cascade above carries this principle from the pedagogical small to the physically large. A co-optimized $\boldsymbol{c}$ is no longer a lens, an antenna, or a SQUID loop that merely transduces; it is a purpose-built information-preserving structure that filters, concentrates, and encodes the incoming signal in the geometry's own complex degrees of freedom, shaped by the task and by the policy that will consume its output. The broader research program we invite is, in this view, a piece of the larger migration of intelligence from digital to physical: in place of a generic measurement whose deficit is rescued by deep networks downstream, the analog structure carries its task-aware share, and the juncture of information loss is relocated, narrowed, and in some regimes transcended. What exact Bellman DP proves possible in the pedagogical small, the relaxed rungs make feasible in the physically large — and the physical structure, once co-optimized, is no longer a conduit for intelligence but a participant in it.

6 Case Study A: Radar beam search (exact DP, discrete state, $\Phi_{\mathbf{MI}}$)

The radar beam-search case is the smallest non-trivial POMDP that exhibits the full V_{oracle} / V_{fixed} / V_{adaptive} / $\mathbb{E}[\text{IG}]$ decomposition, and it serves as the pedagogical entry point for the rest of the paper.

6.1 Problem setup

State. $\boldsymbol{x} \in \{1, \ldots, |\mathcal{X}|\}$ is a target cell on a ring of $|\mathcal{X}| = 16$ cells. Prior $p_0(\boldsymbol{x}) = 1/|\mathcal{X}|$.

Action. At each of $K = 4$ epochs the sensor points a beam at cell $\boldsymbol{s}_k \in \{1, \ldots, |\mathcal{X}|\}$.

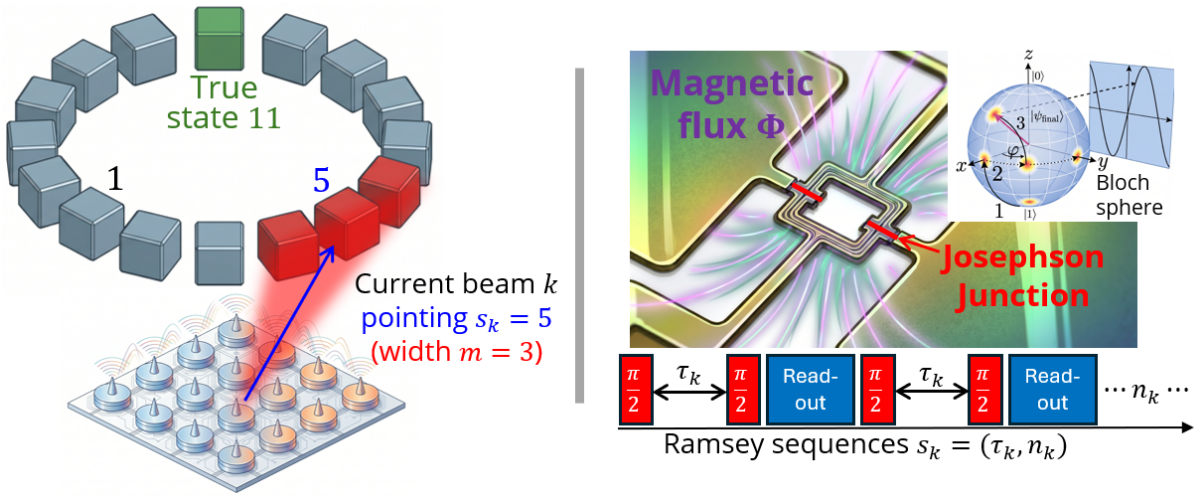

Figure 1: Scene illustrations for the two case studies. *Left* (Case Study A, §6): a phased-array antenna interrogates a ring of 16 target cells; the true state (green, cell 11), the current beam pointing at cell $\boldsymbol{s}_k = 5$ with width $m = 3$, and the adaptive redirect selected by $\pi^\star$ at the next epoch. *Right* (Case Study B, §7): the x-mon transmon cross on sapphire with a SQUID loop (two Josephson junctions) at the end of the $+x$ arm, the external flux Φ_{ext} threading the loop, a meandering readout resonator, and the Bloch-sphere view of the qubit; the bottom timeline shows the adaptive Ramsey sequences $\boldsymbol{s}_k = (\tau_k, n_k)$ chosen by $\pi^\star(\boldsymbol{b}_k)$ at each epoch.

Observation. $\boldsymbol{y}_k \in \{0, 1\}$ (miss/detection) with Bernoulli likelihood

$$p(\boldsymbol{y}_k = 1 \mid \boldsymbol{x}, \boldsymbol{s}_k; \boldsymbol{c}) = p_{\text{FA}} + (p_{\text{D}} - p_{\text{FA}}) \cdot G_{\boldsymbol{c}}(\theta_{\boldsymbol{s}_k} - \varphi_{\boldsymbol{x}}), \tag{27}$$

with $p_{\text{D}} = 0.9$, $p_{\text{FA}} = 0.05$, cell bearings $\varphi_{\boldsymbol{x}} = 2\pi(\boldsymbol{x} - 1)/|\mathcal{X}|$, and pointing angles $\theta_{\boldsymbol{s}_k}$. The geometry family is a top-hat (uniform-sensitivity, rectangular-profile) beam parametrized by integer width $m \in \{1, \ldots, |\mathcal{X}|\}$: the beam centered on cell a covers all cells within a ring distance of $\lfloor m/2 \rfloor$ from a, giving detection probability p_{D} uniformly inside the beam and p_{FA} uniformly outside. m is the discrete design variable we sweep below.

Reward. $\Phi = \Phi_{\text{MI}}$, since the state is discrete and there is no natural MSE.

6.2 Enumeration results

The exact enumeration of V_{oracle}, V_{fixed}, and backward-induction V_{adaptive} at $|\mathcal{X}| = 16$, $K = 4$ is computationally modest (order 10^7 elementary operations, Julia runtime ~ 0.1 s on a single thread). In the narrow-beam ($m = 1$) and wide-beam ($m = 8$) regimes we obtain:

	narrow ($m = 1$)	wide ($m = 8$)
V_{oracle}	2.580 nats	1.923 nats
V_{adaptive}	0.619	1.700
V_{fixed}	0.499	1.411
$\mathbb{E}[\text{IG}]$	2.081	0.513
realized gain $V_{\text{adaptive}} - V_{\text{fixed}}$	0.121	0.289
EVPI saturation	5.8%	56.4%

Note the non-monotone relationship: the narrow regime has the largest $\mathbb{E}[\text{IG}]$ (more than $4\times$ the wide regime's) but the smallest realized adaptive gain (less than half). The narrow EVPI is "loose", most of the adaptive headroom is unreachable by any policy at $K = 4$, while the wide EVPI is relatively tight.

6.3 Sweep over the top-hat beam width

Sweeping the integer beam width $m \in \{1, \ldots, |\mathcal{X}|\}$ and reporting all four functionals (consecutive pairs of m produce identical beams under the integer-width parametrization, so the table lists one representative per degenerate pair; the endpoints $\{1, 2\}$ and $\{15, 16\}$ are both shown to make the pairing explicit):

m	V_{oracle}	V_{adaptive}	V_{fixed}	$\mathbb{E}[\text{IG}]$	realized	sat. %
1	2.580	0.619	0.499	**2.081**	0.121	5.8
2	2.580	0.619	0.499	2.081	0.121	5.8
3	2.293	1.401	1.103	1.190	0.299	25.1
5	2.081	1.631	1.343	0.737	0.287	38.9
7	1.924	1.700	**1.411**	0.513	0.289	56.4
9	1.879	**1.704**	1.401	0.478	**0.303**	**63.5**
11	2.074	1.632	1.320	0.754	0.312	41.4
13	2.312	1.389	1.041	1.271	0.347	27.3
15	2.514	0.611	0.442	2.071	0.169	8.1
16	2.514	0.611	0.442	2.071	0.169	8.1

The optimum for V_{adaptive} is at $m^\star = 9$ ($V^\star_{\text{adaptive}} = 1.704$ nats, saturation 63.5%). V_{fixed} peaks one cell earlier at $m = 7$. Realized adaptive gain $V_{\text{adaptive}} - V_{\text{fixed}}$ peaks at $m = 13$ (0.347 nats). And the ignorance gap $\mathbb{E}[\text{IG}]$ peaks at the two *extremes* $m = 1$ (narrow) and $m = 16$ (isotropic), far from any V_{adaptive} or V_{fixed} maximum.

6.4 Three-way comparison: EVPI picks the wrong geometry

Top-hat three-way comparison.

$\boldsymbol{c}$	$\mathbb{E}[\text{IG}]$	V_{adaptive}	V_{fixed}	realized	sat.
top-hat $m = 1$ (narrow extreme)	**2.081**	0.619	0.499	0.121	5.8%
top-hat $m = 16$ (isotropic extreme)	2.071	0.611	0.442	0.169	8.1%
top-hat $m = 9$ (optimum)	0.478	**1.704**	**1.401**	**0.303**	**63.5%**

The optimum $m^\star = 9$ dominates both extremes on every actionable column: V_{adaptive}, V_{fixed}, realized gain, and saturation. The *only* column on which it is not best is $\mathbb{E}[\text{IG}]$, where both extremes beat it by a factor of four. An EVPI-only geometry selection would reject $m^\star = 9$ in favor of either $m = 1$ or $m = 16$, both of which realize V_{adaptive} that is nearly three times worse. Quantitatively, $V_{\text{adaptive}}(m = 9)/V_{\text{adaptive}}(m = 1) = 1.704/0.619 \approx 2.75$: the EVPI-argmax geometry is strictly suboptimal in attainable adaptive value by a factor of $\sim 2.75\times$ at this problem size.

The same observation underwrites the conceptual claim of Section 3: $\arg\max_{\boldsymbol{c}} \mathbb{E}[\text{IG}](\boldsymbol{c}) \neq \arg\max_{\boldsymbol{c}} V_{\text{adaptive}}(\boldsymbol{c})$, the disagreement is operationally large, and only the full joint solve picks the right geometry.

6.5 Optimal adaptive policy structure

The DP-optimal policy tree reveals structure beyond what the scalar values expose. In the wide ($m = 8$) regime, the depth-2 head of the tree is:

history	action
$\emptyset$	cell 1
$y_1 = 0$ (miss)	cell 4
$y_1 = 1$ (detect)	cell 13

A detection at cell 1 (which covers cells $\{14, 15, 16, 1, 2, 3, 4\}$) narrows the support to the upper ring half; step 2 at cell 13 (covering $\{10, \ldots, 16\}$) bisects that half. This is binary search conditioned on the observation, the hallmark of adaptive value that fixed schedules cannot replicate.

7 Case Study B: Superconducting-qubit flux sensor (exact DP, continuous state, $\Phi_{\mathbf{Var}}$)

Case Study A is discrete-state and has no natural MSE metric; here we take the same exact-DP machinery to a continuous-state problem whose deployment metric is Bayesian MSE, and compare joint-DP co-design against the joint Bayesian Cramér–Rao fixed-schedule baseline that dominates the classical Fisher-information literature for scalar estimation. Joint-DP reduces deployed MSE by 11.3× at $z = +132\sigma$. Figure 1 (right) shows the sensor geometry and the adaptive Ramsey sequences.

7.1 Problem setup

We consider the frequency-tunable transmon flux sensor of Danilin, Nugent & Weides [40]. The unknown is the external magnetic flux $\boldsymbol{x} = \Phi_{\text{ext}}$ threading a SQUID loop, with prior $\boldsymbol{x} \sim \text{Uniform}[0, \phi_{\max}\Phi_0]$ at $\phi_{\max} = 0.1$ (the rationale for this operating point is discussed in §7.6). Ramsey interferometry at a fixed operating flux $\Phi^\star$ returns a Bernoulli outcome with likelihood

$$p(\boldsymbol{y} = 1 \mid \boldsymbol{x}, \tau; \boldsymbol{c}) = \tfrac{1}{2} + \tfrac{1}{2}\,\text{th}(\boldsymbol{x}, \boldsymbol{c})\, e^{-A(\boldsymbol{x},\boldsymbol{c})\tau - B(\boldsymbol{x},\boldsymbol{c})^2\tau^2}\, \cos[\Delta\omega(\boldsymbol{x}, \boldsymbol{c})\tau], \tag{28}$$

with all noise rates (A, B) and spectral coefficients ($\Delta\omega$) closed-form in the 7-dimensional parametric geometry $\boldsymbol{c} = (f_q^{\max}, E_C/h, \kappa, \Delta_{qr}, T, A_\Phi, A_{I_c})$ (see [40] for the complete rate formulas). The action at epoch k is the pair $\boldsymbol{s}_k = (\tau_k, n_k)$, with τ_k the Ramsey delay and n_k the number of repetitions (yielding a Binomial(n_k, p) observation).

Reward. $\Phi = \Phi_{\text{Var}}$, motivated by the Bayesian MSE alignment discussed in Section 2.3.

7.2 Exact Bellman DP via count-statistic memoization

The Ramsey likelihood is invariant under permutation of measurements at the same (τ, n) tuple. Consequently the belief $\boldsymbol{b}_k$ at step k is a sufficient statistic of the cumulative counts $(N_j, M_j)_{j=1}^J$ where N_j is the total number of measurements made at delay τ_j across all past epochs and M_j the count of $\boldsymbol{y} = 1$ outcomes. This makes the belief state a finite tuple of integers, and the Bellman DP admits exact memoization on a count-tuple hash table. At $K = 4$, $J = 10$ (log-spaced τ in $[10, 320]$ ns), $L = 2$ (repetitions $n \in \{1, 10\}$), $K_\Phi = 256$ (belief grid resolution), the memo contains on the order of 10^7 nodes and the DP completes in minutes on a single thread.

7.3 Joint-DP co-design

We run the policy iteration of Section 4 with outer Adam updates at learning rate 5×10^{-4} and periodic policy reoptimization, on a physically realistic restricted geometry box ($f_q^{\max} \in [3, 12]$ GHz, $E_C/h \in [0.15, 0.4]$ GHz, $\kappa \in [0.1, 5]$ MHz, $\Delta_{qr} \in [0.8, 5]$ GHz), with the noise

amplitudes and temperature held fixed at the operating values reported in [40] ($A_\Phi = A_{I_c} = 10^{-6}$, $T = 40$ mK). The outer Adam rescales each $\boldsymbol{c}$ component by its box width before stepping, which is necessary because the raw natural-unit ranges of the 7 components span 14 orders of magnitude. The best $\boldsymbol{c}$ encountered over the trajectory is

$$\boldsymbol{c}_1^\star \approx (f_q^{\max} = 8.99 \text{ GHz},\ E_C/h = 0.254 \text{ GHz},\ \kappa = 0.47 \text{ MHz},\ \Delta_{qr} = 2.02 \text{ GHz}), \tag{29}$$

with $V_{\text{adaptive}}(\boldsymbol{c}_1^\star) = -\mathbb{E}[\text{Var}(\boldsymbol{b}_K)] = -7.49 \times 10^{-5}\,\Phi_0^2$.

7.4 Joint PCRB baseline

The Bayesian Cramér–Rao bound [50, 51] gives an alternative joint optimization:

$$(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star) = \underset{\boldsymbol{c},\boldsymbol{s}_{1:K}}{\arg\max}\ \log J_P(\boldsymbol{s}_{1:K}, \boldsymbol{c}), \qquad J_P = J_0 + \mathbb{E}_{\boldsymbol{x}}\left[\sum_{k=1}^K n_k\, J_F(\boldsymbol{x}, \tau_k, \boldsymbol{c})\right], \tag{30}$$

where $J_F(\boldsymbol{x}, \tau, \boldsymbol{c}) = (\partial p/\partial \boldsymbol{x})^2/[p(1-p)]$ is the per-shot scalar Fisher information. This is a *single-level* optimization over $\boldsymbol{c}$ and a fixed schedule $\boldsymbol{s}_{1:K}$ (no Bellman, no belief tree). Running the analogous outer Adam with inner schedule enumeration at per-iteration reoptimization gives:

$$\boldsymbol{c}_2^\star \approx (f_q^{\max} = 9.62 \text{ GHz},\ E_C/h = 0.272 \text{ GHz},\ \kappa = 0.10 \text{ MHz},\ \Delta_{qr} = 3.20 \text{ GHz}), \tag{31}$$

with $\boldsymbol{s}_2^\star = (\tau_{10}, n_2)^4 = (320 \text{ ns}, 10)^4$. The PCRB-optimal geometry $\boldsymbol{c}_2^\star$ prefers a larger qubit frequency, a larger qubit-resonator detuning, and the minimum-feasible κ, whereas the joint-DP optimum $\boldsymbol{c}_1^\star$ settles closer to the experimental nominal value of each. The two optima differ materially inside the restricted box, so the deployed-MSE gap reported below is a full joint co-design comparison rather than a shared-geometry test of policy structure alone.

7.5 Deployed MSE comparison

We deploy each design with $n_{\text{mc}} = 20{,}000$ Monte-Carlo trajectories at $K_\Phi = 256$ posterior resolution, paired common-seed $\boldsymbol{x}_{\text{true}}$ draws. Each deployment uses the posterior-mean estimator. Results:

quantity	value
$\overline{\text{MSE}}_1$ (joint DP)	$(7.49 \pm 0.24) \times 10^{-5}\ \Phi_0^2$
$\overline{\text{MSE}}_2$ (PCRB baseline)	$(8.43 \pm 0.05) \times 10^{-4}\ \Phi_0^2$
$J_P^{-1}(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ (CRB)	$1.03 \times 10^{-9}\ \Phi_0^2$
ratio $\overline{\text{MSE}}_2/\overline{\text{MSE}}_1$	**11.27**
z-score	**+132σ**

Joint-DP $(\boldsymbol{c}_1^\star, \pi_1^\star)$ beats the PCRB baseline $(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ by 11.3× on deployed Bayesian MSE at $K = 4$, at $z = +132\sigma$.

The CRB lower bound $J_P^{-1} \approx 10^{-9}$ is five-to-six orders of magnitude below both empirical MSEs, a reminder that the asymptotic CRB is loose for finite-horizon biased posterior-mean estimators on bounded-support priors [76, 103]. Neither design approaches the CRB lower bound; the comparison between the two is nevertheless clean and statistically decisive.

7.6 Why $\phi_{\max} = 0.1$: aliasing, horizon, and PCRB validity

The prior width $\phi_{\max}$ interacts with the tractable horizon $K = 4$ through Ramsey aliasing. The likelihood (28) is periodic in $\Delta\omega(\boldsymbol{x})\,\tau$, so a prior that spans many Ramsey fringes yields a multimodal posterior that can be disambiguated only by measurements at sufficiently varied τ — to leading order, at least $\log_2(N_{\text{fringes}})$ distinct delays are required. At $\phi_{\max} \to \Phi_0/2$ (uninformative prior) the support spans many fringes and four epochs are too few for either a fixed or an adaptive schedule to resolve them; both joint-DP and joint-PCRB converge to comparable aliasing-floor MSE and the adaptive margin shrinks. At $\phi_{\max}$ well below this the prior is narrow enough that aliasing is essentially absent, both methods operate far below the floor, and the joint-DP margin grows rapidly. Separately, the PCRB is a local/asymptotic bound that elides posterior multimodality, so at large $\phi_{\max}$ it is mis-specified as a baseline: the comparison would be doubly uninformative (joint-DP K-limited, joint-PCRB outside its regime of validity). Choosing $\phi_{\max} = 0.1$ sits just inside the regime where joint-DP's adaptive finesse becomes observable at $K = 4$ while keeping the PCRB baseline a valid reference — a representative rather than extremal test of co-design quality.

8 Case Study C: Photonic metasensor (Gaussian-trace surrogate, 10^5-dim continuous $\boldsymbol{c}$, $\Phi_{\log J}$)

The two preceding case studies used exact Bellman DP. The photonic case study operates on a $\sim 10^5$-pixel permittivity landscape where exact DP is infeasible, and demonstrates the relaxation-hierarchy rung 4: Gaussian-posterior belief updates via an extended Kalman filter, the Bayesian Fisher-information-matrix reward $\Phi_{\log J}$, and implicit-function-theorem differentiation through the inner optimization that selects the adaptive probe phases.

8.1 Problem setup

The device is a 2×2 lattice of identical 2D cross-waveguide scatterers, each cell a four-port cross with a central 300×300-pixel permittivity design region; all four cells share the same inverse-designed geometry, giving $\boldsymbol{c} \in [0,1]^{300\times 300}$ (equivalently $\boldsymbol{c} \in \mathbb{R}^{90{,}000}$) as the outer optimization variable. The unknown state $\boldsymbol{x} = \mathrm{vec}(\Delta n) \in \mathbb{R}^4$ is the vector of per-cell refractive-index perturbations to be estimated. At each of $K_{\text{train}} = 3$ measurement epochs during co-design the sensor chooses a pair of input phases $\boldsymbol{s}_k = (\phi_1^{(k)}, \phi_2^{(k)})$ applied to two of the lattice's external input ports as coherent broadband drives across 20 frequencies; the observations are the $\boldsymbol{y}_k \in \mathbb{R}^8$ broadband port powers measured at the lattice's 8 external output ports, with additive Gaussian noise.

Reward. $\Phi = \Phi_{\log J}$ with J_N the Bayesian Fisher-information matrix accumulated over the K_{train} epochs. The Gaussian-posterior Kalman filter makes $\log\det J_N$ tractable in closed form at each epoch.

8.2 Physics and differentiability

The forward model is a 2D finite-difference frequency-domain (FDFD) Maxwell solver applied to a single unit cell, which returns one 4×4 S-matrix per frequency on a broadband sampling grid of 20 frequencies. We Taylor-expand the per-cell S-matrix to second order in the local refractive-index perturbation Δn,

$$S(\Delta n) \approx S_0 + \frac{\partial S}{\partial n}\Delta n + \frac{1}{2}\frac{\partial^2 S}{\partial n^2}\Delta n^2, \tag{32}$$

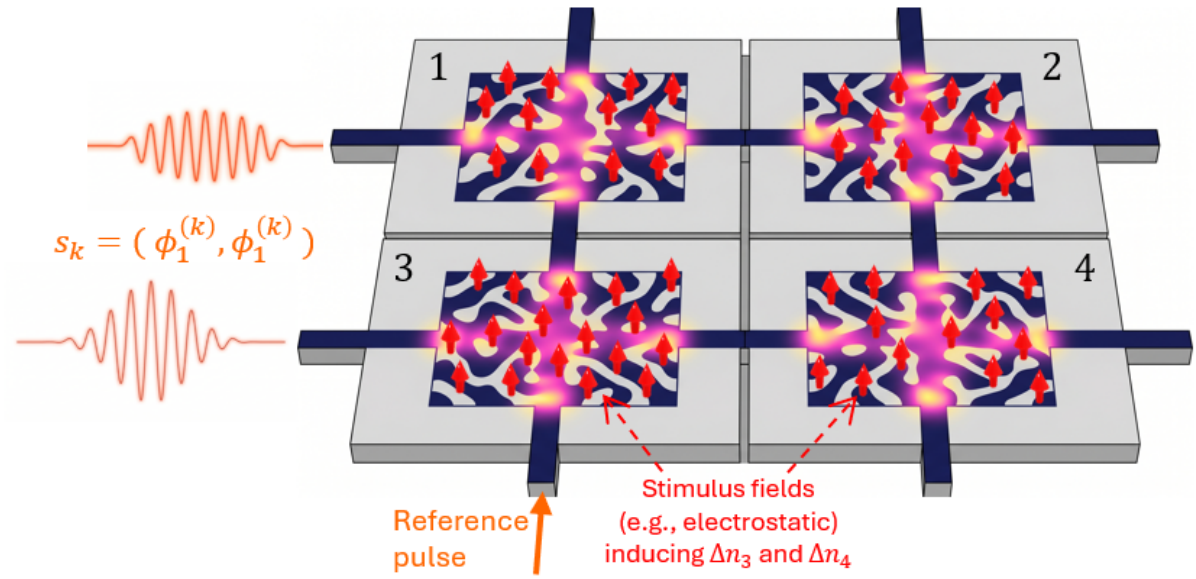

Figure 2: Case Study C overview. A 2×2 lattice of identical cross-waveguide scatterers (cells 1–4), each carrying the same inverse-designed binary-permittivity pattern in its central design region. Two coherent broadband light pulses enter at the left edge: an action pulse carrying the adaptive input phases $\boldsymbol{s}_k = (\phi_1^{(k)}, \phi_2^{(k)})$ at epoch k, and a fixed-phase reference pulse. External stimulus fields (e.g., electrostatic) acting on the cells induce per-cell refractive-index shifts $\Delta n_1, \ldots, \Delta n_4$, which constitute the hidden state $\boldsymbol{x} = \mathrm{vec}(\Delta n) \in \mathbb{R}^4$ to be estimated. The intensity overlays inside each cell visualize the electromagnetic near-field response under one such probe. Port powers measured at the 8 external output ports form the observation $\boldsymbol{y}_k \in \mathbb{R}^8$; an EKF reconstructs $\boldsymbol{x}$ across successive broadband measurement steps, with the adaptive policy re-selecting $\boldsymbol{s}_{k+1}$ from $\boldsymbol{x}$'s posterior after each step.

computed with a single FDFD solve per frequency and evaluated independently at each of the four cells with its own Δn_i. The four cells are wired together by the port-connection equations to assemble the lattice's 8-port S-matrix; the 8 external-port powers, summed over the 20 frequencies against a Gaussian spectral weight, form the observation $\boldsymbol{y} \in \mathbb{R}^8$. The Jacobian $\partial \boldsymbol{y}/\partial \boldsymbol{c}$ passes through the cell S-matrix coefficients via the FDFD adjoint method [88, 89].

8.3 Joint optimization

The outer loop is the policy iteration of Section 4. At each $\boldsymbol{c}$ the inner optimization $\boldsymbol{s}^\star(\boldsymbol{b}_k; \boldsymbol{c}) = \arg\max_{\boldsymbol{s}} \Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ is solved by grid search followed by L-BFGS refinement. Its derivative with respect to $\boldsymbol{c}$, which the chain rule requires, is computed in closed form from the implicit function theorem applied at the inner argmax, avoiding nested automatic differentiation. To produce a geometry compatible with lithographic fabrication we apply a density filter to $\boldsymbol{c}$ (convolving with a radius-R kernel to enforce a minimum feature size) and then a tanh threshold projection with sharpness parameter β [104–107]. During optimization β is increased along a continuation schedule $16 \to 32 \to 64 \to 128 \to 256$ so that the projection gradually sharpens from a smooth blur to a near-binary mask.

8.4 Optimized binary design and loss trajectory

8.5 Monte-Carlo EKF deployment

Deploying the optimized binary design with the EKF reconstruction and fresh perturbation draws (200 episodes, $K_{\text{deploy}} = 10$ EKF steps per episode):

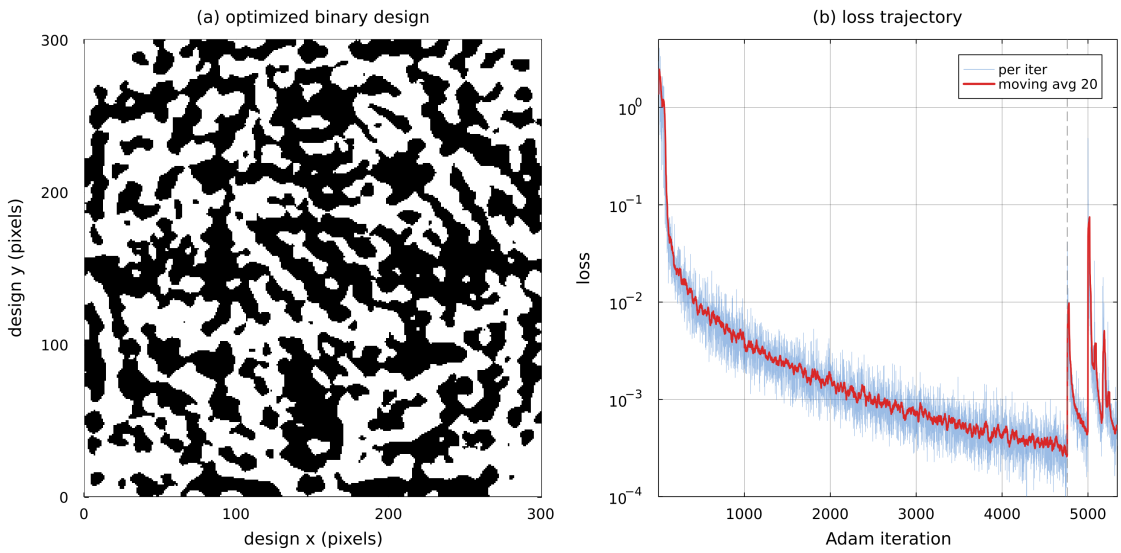

Figure 3: Final optimized binary permittivity of the 300×300 design region (main panel). Inset: joint-DP loss trajectory from a fully random initialization through the full co-design; light blue, per-iteration loss; red, 20-iteration moving average; dashed, the transition into the filter-constrained projection phase.

metric	optimized	random initial	ratio
EKF relative error (10 steps)	0.97%	120%	**123**× better
BFIM trace (step 1)	2.50×10^{13}	4.01×10^{10}	**622**×
Episode loss (10 steps)	3.5×10^{-4}	6.5	$\mathbf{1.84 \times 10^4}$×

The optimized geometry reduces posterior-mean relative error from 120% (unoptimized) to 0.97% (optimized) after ten EKF measurement steps, a 123× improvement, while delivering 622× more Fisher information per measurement. After a single broadband measurement the EKF reduces relative error from 184% (prior) to 3.4% (posterior), essentially solving the estimation problem with one observation. These ratios are enormous because the problem starts ill-conditioned: a random permittivity distribution produces a nearly non-invertible S-matrix. The improvement illustrates what high-dimensional $\boldsymbol{c}$-optimization buys when jointly designed with the EKF reconstruction.

8.6 Feasibility discussion

At the design frequency $\lambda = 1550$ nm, the shot-noise-limited detection threshold we report corresponds to $\sim 64\,\mu$W per port at $1\,\mu$s integration, within the operating range of standard telecom lasers. The final binary permittivity distribution is smooth at the density-filter length scale ($R = 5$ pixels ≈ 200 nm feature size at 5-nm/pixel lattice) and compatible with electron-beam lithography on standard SOI [108, 109].

9 Discussion

9.1 The common shape of the adaptive advantage

Across the three case studies the adaptive policy's margin over the best fixed-schedule baseline comes from one move: redirecting the next measurement based on whichever partition of the state space the current data has already ruled out. In Case A, a detection at cell 1 halves the ring's support before step 2 targets cell 13. In Case B, the first epoch at $\tau \approx 22$ ns collapses the

multimodal posterior before later epochs switch to $\tau = 320$ ns to sharpen within a single fringe; the PCRB fixed schedule commits to $\tau = 320$ ns from step 1 and cannot undo the aliasing. In Case C, each new pair of probe phases is re-aligned with the largest remaining eigenvector of the EKF posterior covariance, so that successive measurements shrink the least-constrained direction of the state. Prior-averaged information criteria, whether mutual information, Fisher information, or EVPI, marginalize over the data before committing an action, so the geometry they recommend is tuned to whichever schedule is best on average rather than best conditionally. The joint-DP margin is therefore not a numerical artifact of any particular problem: it is the signature of a hardware operating point chosen in anticipation of a conditional policy, and hardware chosen without that anticipation cannot, in general, recover it by attaching an adaptive policy after the fact.

9.2 Why the PCRB baseline loses even when it co-optimizes $\boldsymbol{c}$

The 11.3× Case B margin sharpens this general statement. Both $\boldsymbol{c}_1^\star$ and $\boldsymbol{c}_2^\star$ are genuine numerical optima of their respective objectives inside the same physical box; nothing prevents the PCRB optimizer from landing on $\boldsymbol{c}_1^\star$ if that geometry gave it more Fisher information. It lands on a different point $\boldsymbol{c}_2^\star$ and still produces MSE an order of magnitude worse, because the two objectives reward different physical behavior. The Fisher information at long Ramsey delay is high because one long measurement gives sharp local phase resolution, but at prior width $\phi_{\max} = 0.1\,\Phi_0$ the prior support spans many Ramsey fringes and the single-level Fisher criterion has no machinery for disambiguating which fringe the true Φ_{ext} lies on. The Bellman-optimal policy does have that machinery: a short-τ first epoch reduces the multimodal posterior to a unimodal one, and the remaining epochs exploit long-τ resolution within the selected fringe. The PCRB-optimal $\boldsymbol{c}$ is therefore tuned to a schedule the adaptive policy would never deploy; and because the coherence and drive parameters that favor long-τ fringe visibility differ from those that favor short-τ aliasing-breaking, the two optimizers converge to different geometries entirely. The belief-conditional action selection is not a refinement one can bolt on afterwards, which is what makes co-design, rather than serial "pick $\boldsymbol{c}$ then pick π" optimization, the minimum principled procedure.

9.3 What the EVPI *is* still good for

We advocate demoting the EVPI from a design target to a workflow tool. Three concrete roles survive:

1. A pre-screen when the full joint DP is computationally prohibitive but schedule enumeration remains tractable. $\mathbb{E}[\text{IG}](\boldsymbol{c})$ requires only V_{oracle} and V_{fixed}, both computable by direct enumeration of candidate schedules, bypassing the backward-in-time value iteration that V_{adaptive} demands; the cheapness of EVPI is inherited from the cheapness of those enumerations (small discrete action sets or low-dimensional continuous actions on a reasonable grid), and evaporates when the schedule space is large enough that enumeration itself becomes the bottleneck.
2. A post-hoc diagnostic showing how much adaptive headroom is available at the joint-optimum geometry. Small $\mathbb{E}[\text{IG}](\boldsymbol{c}^\star)$ means the joint-optimum is close to the oracle ceiling; large values indicate remaining value from longer horizons, richer action sets, or tighter priors.
3. An interpretive lens on the shape of the $\boldsymbol{c}$-landscape. The non-coincidence of $\arg\max \mathbb{E}[\text{IG}]$ and $\arg\max V_{\text{adaptive}}$ encodes physics: extreme geometries have large ceilings that no policy can reach; interior geometries have smaller ceilings that adaptive policies can consume a larger fraction of.

9.4 Limitations

- **Horizon.** Case Studies A and B are at small horizon ($K = 4$) where exact DP is tractable. At larger horizons the Bellman memoization table blows up, forcing descent down rungs 2–4 of the hierarchy. The Gaussian-posterior EKF surrogate (rung 4) extrapolates to arbitrary K; its accuracy relative to exact DP at moderate K is an open question.
- **Parametric $\boldsymbol{c}$ in Case B.** Case Study B's 7-dimensional geometry is the closed-form-parametrized subset of the transmon circuit, not the full layout design space. Extending $\boldsymbol{c}$ to include SQUID-loop mutual inductances via analytically differentiable Biot–Savart [110] is architecturally supported and expected to give joint-DP further advantage; this is ongoing work.
- **Finite sample MSE bound.** The Bayesian CRB $1/J_P$ is an asymptotic lower bound. For biased Bayesian estimators on bounded priors at small K the bound is loose by orders of magnitude (Case Study B: $\sim 10^5$–$10^6\times$ gap between the CRB and the deployed MSE of either design). Tighter finite-sample bounds such as the Weiss–Weinstein [111], Bobrovsky–Zakai, and Ziv–Zakai [77] bounds would be more informative anchors; integrating them into the framework is a natural next step.
- **Continuous-action Bellman.** The count-statistic memoization exploited in Case Study B breaks for continuous actions. Practical paths include fine discretization (as we did), Gaussian belief approximations (rung 4), or particle-filter DP. A full continuous-action Bellman for the scqubit problem at $K \geq 4$ would require one of these paths; we presented the discretization path as the cheapest.

9.5 Open directions: dynamical state

All three case studies treat the unknown $\boldsymbol{x}$ as static over the K-step measurement horizon. The natural next frontier is *dynamical* $\boldsymbol{x}$: targets that move, circuit parameters that drift, fields that fluctuate across the sensor integration window. Three things change when the state evolves. First, the belief filter becomes a recursive predict-correct step rather than a single-shot posterior update, and for non-Gaussian posteriors the inference primitive has to move from grids and EKFs to particle filters. Second, the Bellman backup has to anticipate state transitions over the K steps, not just measurement accumulation, so the dynamics time scale interacts with the horizon and with each rung's coarsening of the belief summary. Third, when the action space is high-dimensional or continuous (beam angles, integration times, waveform parameters) the prescribed closed-form $\arg\max$ of §5 becomes unwieldy and a neural-network policy becomes the natural inner solver, which re-introduces the hardware-light pathology flagged at the branch point of the cascade. The envelope-theorem outer gradient survives this transition unchanged; what does not survive is the exactness of the inner solve, and the clean cascade of rungs 1–4 is replaced by a spectrum of training regimes whose geometry gradients depend on policy capacity, dynamics time scale, and filter fidelity together.

We pursue this extension in a companion line of work on real-time tracking of a nonlinear dynamical system with a jointly-designed physical sensor and a learned measurement policy; full treatment is deferred to a future paper. The present work establishes the static-state foundation on which that extension stands.

10 Conclusion

For any sensor whose physical hardware is designed once but whose measurement policy runs for the lifetime of the device, programmable metasurfaces, cognitive radars, frequency-tunable quantum sensors, reconfigurable photonic-integrated meshes, joint optimization of geometry and policy is the minimum principled procedure. The envelope theorem makes the joint optimization tractable; a principled relaxation hierarchy makes it portable across scales; a family

of compatible scalar rewards makes it applicable across communities. The Expected Value of Perfect Information, classical and instructive, serves as a diagnostic: its argmax disagrees with the joint-DP argmax by factors of $\sim 3\times$ in attainable adaptive value. The deployment-aligned comparison of joint-DP and Bayesian-CRB co-design on a superconducting-qubit flux sensor gives the operational headline: an $11.3\times$ reduction in Bayesian MSE at $z = +132\sigma$, with both hardware geometries numerically co-optimized under their respective objectives.

Code and data. The Julia implementation of all three case studies, including the exact Bellman DP with count-statistic memoization (Case Studies A and B), the envelope-theorem-based outer Adam with per-component box-width preconditioning, the MSE-terminal reward (`Bellman.jl`, `Gradient.jl`, `JointOpt.jl`), the joint PCRB baseline (`PCRB.jl`), and the photonic BFIM+EKF surrogate (`SimGeomBroadBand.jl`, `BFIMGaussian.jl`, `PhEnd2End.jl`), is publicly available. Numerical outputs of all tables in Sections 6–8 are serialized and shipped with the source; reproducing any table requires a single Julia invocation.

Acknowledgments. [redacted for blind review; funding, collaborators, and facility acknowledgments here.]

A Derivation of the Bellman recursion from $V(\boldsymbol{c}, \pi)$

This appendix derives Equations (4)–(5) from Equation (2), isolating the three ingredients — change of conditioning, sufficiency of the posterior, and Bellman's principle of optimality — that together produce the reduction.

Step 1: Change of conditioning. Equation (2) writes the objective as a nested expectation over the prior and the data trajectory. Swap the order using the law of total expectation — condition on the data rather than on the state:

$$V(\boldsymbol{c}, \pi) \;=\; \mathbb{E}_{\boldsymbol{y}_{1:K}|\pi,\boldsymbol{c}}\big[\, \mathbb{E}_{\boldsymbol{x}|\boldsymbol{y}_{1:K},\boldsymbol{c}}\big[\Phi(\boldsymbol{x}, \boldsymbol{s}^{\pi}_{1:K}, \boldsymbol{c})\big] \,\big] \,. \tag{33}$$

The inner expectation conditions on the complete measurement history.

Step 2: Sufficiency of the posterior. The terminal posterior

$$\boldsymbol{b}_K(\boldsymbol{x}) \;\coloneqq\; p(\boldsymbol{x} \mid \boldsymbol{y}_{1:K}, \boldsymbol{s}_{1:K}, \boldsymbol{c}) \tag{34}$$

is a sufficient statistic of the history for inference about $\boldsymbol{x}$: every expectation of a function of $\boldsymbol{x}$ given the history may be re-expressed as an expectation given the posterior,

$$\mathbb{E}_{\boldsymbol{x}|\boldsymbol{y}_{1:K},\boldsymbol{c}}\big[\Phi(\boldsymbol{x}, \boldsymbol{s}^{\pi}_{1:K}, \boldsymbol{c})\big] \;=\; \mathbb{E}_{\boldsymbol{x}|\boldsymbol{b}_K}\big[\Phi_0(\boldsymbol{x}, \boldsymbol{b}_K;\, \boldsymbol{c})\big], \tag{35}$$

where Φ_0 is the pointwise reward re-indexed by the terminal belief. For Φ_{Var} and Φ_{MI} the schedule dependence drops out once the posterior is fixed; for $\Phi_{\log J}$ the belief is augmented to carry the accumulated Fisher matrix so that schedule-dependent accumulators ride along with the posterior (see §2.3).

Step 3: Trajectory factorization via the belief update. Any data trajectory $\boldsymbol{y}_{1:K}$ can be drawn one step at a time by iterating the Bayesian belief update T (Equation (6)), with each new observation sampled conditional on the current belief and action. Applied to the trajectory expectation in Equation (33), this gives, for any fixed policy π, the *policy-evaluation* recursion

$$U^{\pi}_K(\boldsymbol{b};\, \boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{x}|\boldsymbol{b}}\big[\Phi_0(\boldsymbol{x}, \boldsymbol{b};\, \boldsymbol{c})\big], \tag{36}$$

$$U^{\pi}_k(\boldsymbol{b};\, \boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{y}_k|\boldsymbol{b},\pi(\boldsymbol{b}),\boldsymbol{c}}\big[U^{\pi}_{k+1}\big(T(\boldsymbol{b}, \pi(\boldsymbol{b}), \boldsymbol{y}_k;\, \boldsymbol{c});\, \boldsymbol{c}\big)\big], \quad k = K-1, \ldots, 0, \tag{37}$$

with $V(\boldsymbol{c}, \pi) = U_0^\pi(p_0;\, \boldsymbol{c})$. This is Equation (2) rewritten as a backward recursion — no optimization yet, just Bayesian bookkeeping.

Step 4: Bellman's principle of optimality. Define $V_k(\boldsymbol{b};\, \boldsymbol{c}) := \max_\pi U_k^\pi(\boldsymbol{b};\, \boldsymbol{c})$. Maximizing over π factors step-by-step. At step k, parametrize the policy by the action it prescribes at the current belief, $\boldsymbol{s}_k = \pi(\boldsymbol{b})$, and its continuation π' over steps $\{k+1, \ldots, K-1\}$. Equation (37) depends on these two pieces separately: the current action $\boldsymbol{s}_k$ enters the one-step measurement distribution, and the continuation enters via $U_{k+1}^{\pi'}$. Maximizing over both gives

$$V_k(\boldsymbol{b};\, \boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_k \in \mathcal{S}} \; \max_{\pi'} \; \mathbb{E}_{\boldsymbol{y}_k | \boldsymbol{b}, \boldsymbol{s}_k, \boldsymbol{c}}\big[U_{k+1}^{\pi'}\big(T(\boldsymbol{b}, \boldsymbol{s}_k, \boldsymbol{y}_k;\, \boldsymbol{c});\, \boldsymbol{c}\big)\big]. \tag{38}$$

The inner $\max_{\pi'}$ may be pushed inside $\mathbb{E}_{\boldsymbol{y}_k}$: adaptive continuations are history-dependent, so the maximizer π' can itself depend on the realization of $\boldsymbol{y}_k$, and maximizing a family of $\boldsymbol{y}_k$-dependent continuations pointwise before averaging gives the same result as maximizing their average (the interchange of max and $\mathbb{E}$ over $\boldsymbol{y}_k$-measurable policies). Applying this interchange and substituting the definition of V_{k+1},

$$V_k(\boldsymbol{b};\, \boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_k \in \mathcal{S}} \; \mathbb{E}_{\boldsymbol{y}_k | \boldsymbol{b}, \boldsymbol{s}_k, \boldsymbol{c}}\big[V_{k+1}\big(T(\boldsymbol{b}, \boldsymbol{s}_k, \boldsymbol{y}_k;\, \boldsymbol{c});\, \boldsymbol{c}\big)\big], \tag{39}$$

which is Equation (5). The Bellman-optimal policy is read off at each step as

$$\pi_k^\star(\boldsymbol{b};\, \boldsymbol{c}) \;=\; \arg\max_{\boldsymbol{s}_k \in \mathcal{S}} \; \mathbb{E}_{\boldsymbol{y}_k | \boldsymbol{b}, \boldsymbol{s}_k, \boldsymbol{c}}\big[V_{k+1}\big(T(\boldsymbol{b}, \boldsymbol{s}_k, \boldsymbol{y}_k;\, \boldsymbol{c});\, \boldsymbol{c}\big)\big], \tag{40}$$

and evaluating the recursion at the root belief $\boldsymbol{b}_0 = p_0$ recovers $V^\star(\boldsymbol{c}) = \max_\pi V(\boldsymbol{c}, \pi) = V_0(p_0;\, \boldsymbol{c})$ of Equation (7).

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